

Identify the vertices, foci, asymptotes, direction of opening, length of the transverse axis, length of the conjugate axis, length of the latus rectum, and eccentricity of each.

7) $x^2 - y^2 - 6x + 16y - 119 = 0$

8) $-4x^2 + y^2 + 8x - 20y - 4 = 0$

Use the information provided to write the standard form equation of each hyperbola.

9) Vertices: $(1 + 3\sqrt{15}, -7), (1 - 3\sqrt{15}, -7)$
Endpoints of Conjugate Axis: $(1, -7 + 5\sqrt{5})$
 $(1, -7 - 5\sqrt{5})$

10) Vertices: $(19, 2), (1, 2)$
Foci: $(10 + \sqrt{130}, 2), (10 - \sqrt{130}, 2)$

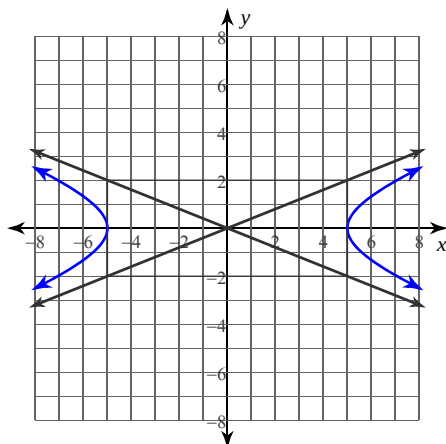
11) Vertices: $(-10, 1), (-10, -17)$
Perimeter of Central Rectangle = 76

12) Vertices: $(7, -2), (5, -2)$
Asymptotes: $y = 11x - 68$
 $y = -11x + 64$

13) Center at $(10, -4)$
Transverse axis is vertical and 18 units long
Conjugate axis is 10 units long

14) Foci: $(6, 1 + 2\sqrt{58}), (6, 1 - 2\sqrt{58})$
Points on the hyperbola are 28 units closer to one focus than the other

15)



16) Center at $(-6, 9)$
Vertex at $(-18, 9)$
Eccentricity = $\frac{5}{4}$